

Github repository: <https://github.com/qssdev20/Sprint0.git>

Sprint #0 Report

Instructions

Objectives

- Make decisions on the Solitaire software development project.
- Learn unit testing and GUI programming in the language of your choice.

Deliverables and Grading Policy

Read the "CS 449 Homework Overview" document **carefully** and make the key decisions for the software development. Use the following template to complete your report.

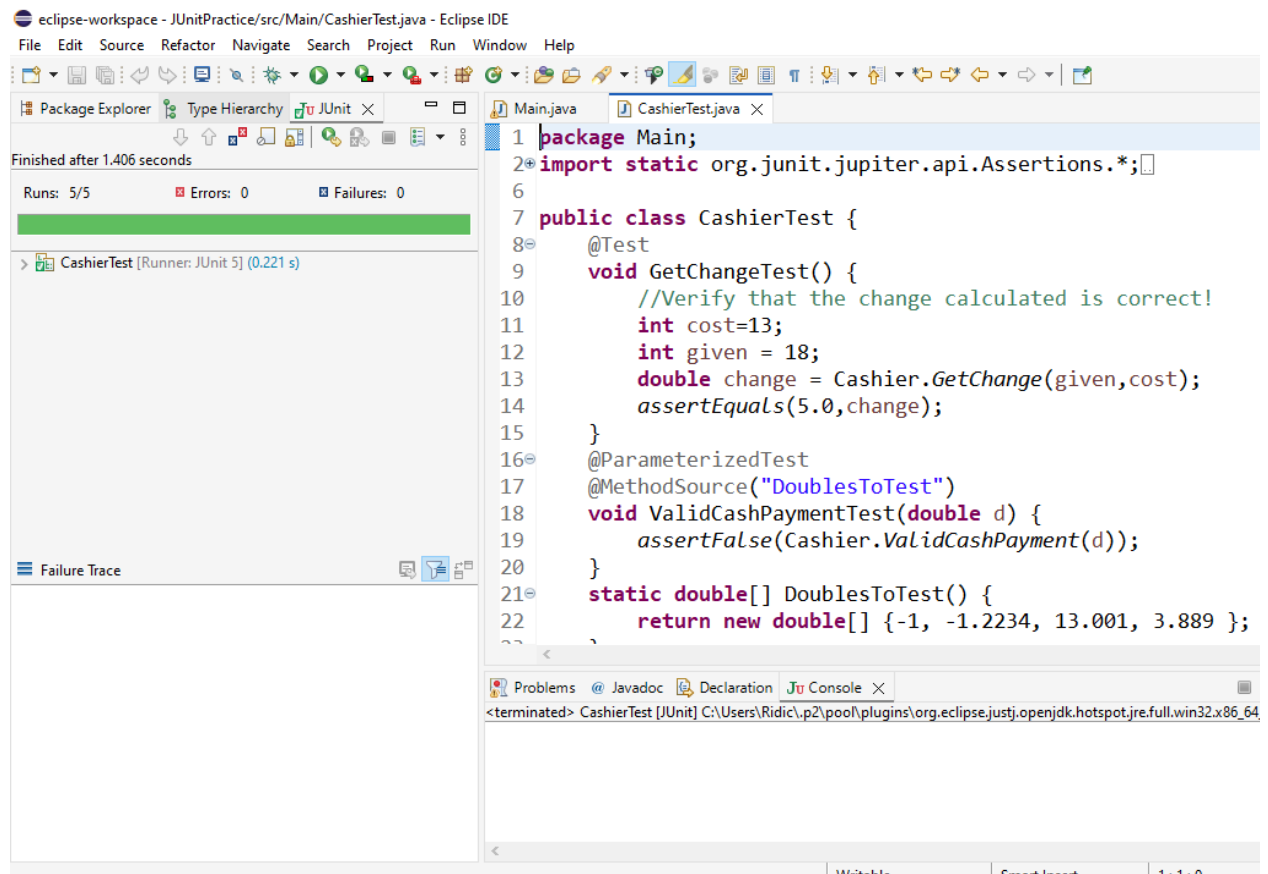
1. Key Decisions of the Solitaire Project (2 points)

Object-oriented programming language (if not using OOP, then justify the use of the language)	Java
GUI library (strongly encouraged)	Swing
IDE (Integrated Development Environment)	Eclipse
xUnit framework (e.g., JUnit for Java)	JUnit
Programming style guide (must read it carefully)	Java
Project hosting site	Github.com
Other decisions if applicable	

Sample programming style guides:

- Google Java Style Guide: <https://google.github.io/styleguide/javaguide.html>
- Google C++ Style Guide: <https://google.github.io/styleguide/cppguide.html>
- Google Python Style Guide: <https://google.github.io/styleguide/pyguide.html>

Unit testing (4 points) Find a tutorial on the unit test framework you have chosen and write at least two xUnit tests of a program you have written or found elsewhere. Attach here (1) the screenshot of your program execution and (2) the source code of your program. For example, Junit5 user guide: <https://docs.junit.org/5.10.3/user-guide/index.html> 3.



Source Code:

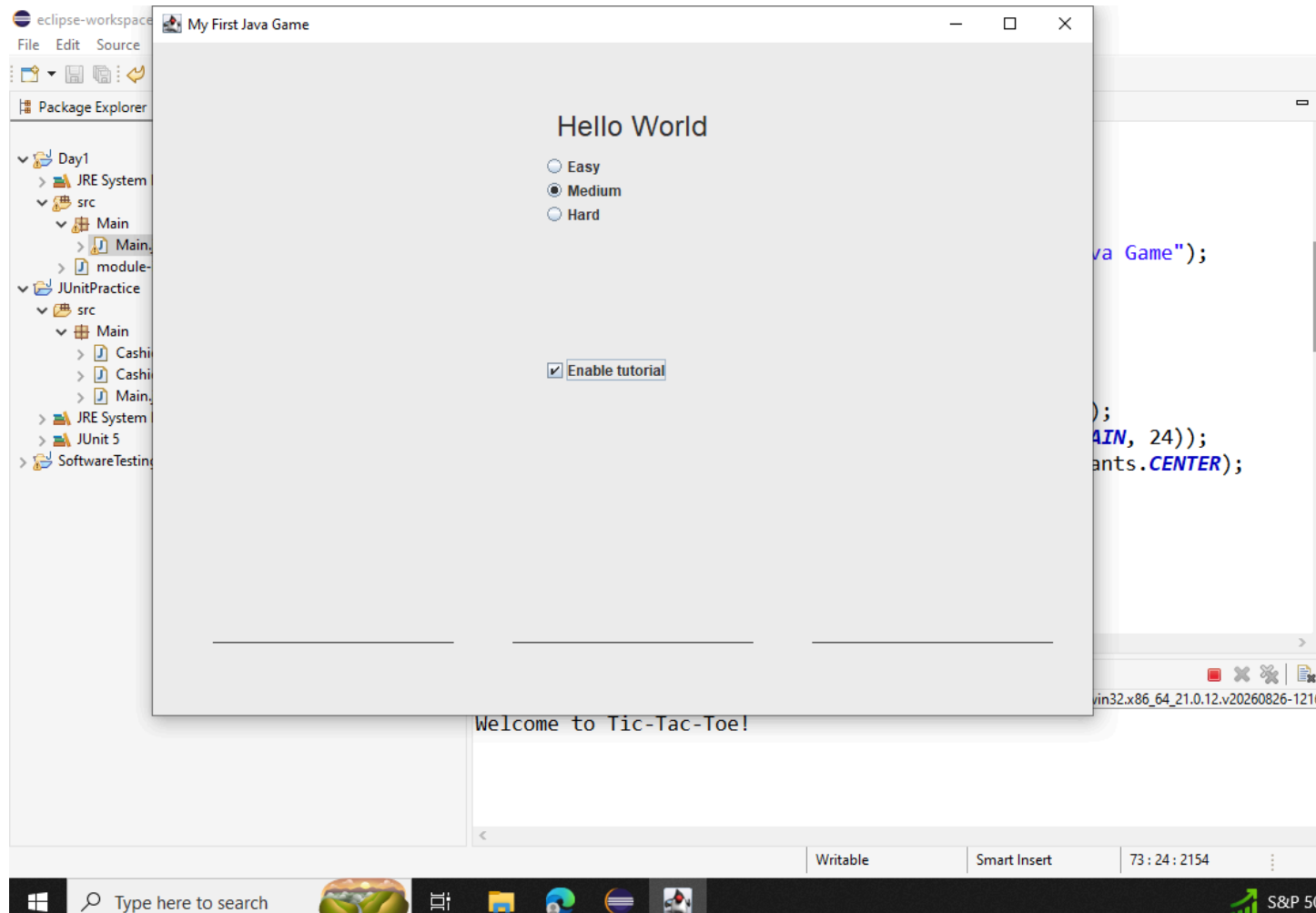
```

1 package Main;
2 import static org.junit.jupiter.api.Assertions.*;
6
7 public class CashierTest {
8     @Test
9     void GetChangeTest() {
10         //Verify that the change calculated is correct!
11         int cost=13;
12         int given = 18;
13         double change = Cashier.GetChange(given,cost);
14         assertEquals(5.0,change);
15     }
16     @ParameterizedTest
17     @MethodSource("DoublesToTest")
18     void ValidCashPaymentTest(double d) {
19         assertFalse(Cashier.ValidCashPayment(d));
20     }
21     static double[] DoublesToTest() {
22         return new double[] {-1, -1.2234, 13.001, 3.889 };
23     }
24 }

```

GUI programming (4 points) Write a GUI program in the language you have chosen for your Solitaire project. The GUI of your program must include text, lines, a check box, and radio buttons. In this assignment, any GUI program of your own work is acceptable. Attach here (1) the screenshot of your program execution and (2) the source code of your program.

Running Code:



Source code:

```
package Main;
```

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
public class Main {
```

```
    public static class Line extends JPanel {
```

```
        @Override
```

```
        protected void paintComponent(Graphics g) {
            super.paintComponent(g);
```

```
            Graphics2D g2 = (Graphics2D) g;
```

```
            //Line 1
```

```
            g2.drawLine(50, 500, 250, 500);
```

```

        g2.drawLine(300, 500, 500, 500);
        g2.drawLine(550, 500, 750, 500);
    }
}

public static void main(String[] args) {

    JFrame window = new JFrame("My First Java Game");

    window.setSize(800, 600);
    window.setLayout(null);

    // Create text
    JLabel label = new JLabel("Hello World");
    label.setFont(new Font("Arial", Font.PLAIN, 24));
    label.setHorizontalAlignment(SwingConstants.CENTER);

    // x, y, width, height
    label.setBounds(250, 50, 300, 40);

    window.add(label);

    // Create drawing panel
    Line line = new Line();
    line.setBounds(0, 0, 800, 600);
    window.add(line);

    //Add a checkbox
    JCheckBox checkBox = new JCheckBox("Enable tutorial");

    //Checkbox panel x,y,width, height
    checkBox.setBounds(325, 260, 125, 26);

    window.add(checkBox);

    //Add to delegate, the method to call when the checkbox
    //state has changed
    checkBox.addActionListener(e -> {
        if (checkBox.isSelected()) {
            System.out.println("Welcome to Tic-Tac-Toe!");
        } else {
            System.out.println("Tutorial OFF");
        }
    });
}

```

```
//Radio button1
JRadioButton easyButton = new JRadioButton("Easy");
JRadioButton mediumButton = new JRadioButton("Medium");
JRadioButton hardButton = new JRadioButton("Hard");

ButtonGroup buttonGroup = new ButtonGroup();
buttonGroup.add(easyButton);
buttonGroup.add(mediumButton);
buttonGroup.add(hardButton);

easyButton.setBounds(325, 90,125, 26);
mediumButton.setBounds(325, 110,125, 26);
hardButton.setBounds(325, 130,125, 26);

window.add(easyButton);
window.add(mediumButton);
window.add(hardButton);

window.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
window.setLocationRelativeTo(null);
window.setVisible(true);
    }
}
```